

■ 3.3 Polynomials and Rational Functions

■ 3.3.1 Structural Operations on Polynomials

<code>Expand[poly]</code>	expand out products and powers
<code>Factor[poly]</code>	factor completely
<code>FactorTerms[poly]</code>	pull out common factors that appear in each term
<code>Collect[poly, x]</code>	arrange a polynomial as a sum of powers of x
<code>Collect[poly, {x, y, ...}]</code>	arrange a polynomial as a sum of powers of x, y, ...

Structural operations on polynomials.

Here is a polynomial in one variable.

```
In[1]:= (2 + 4 x^2)^2 (1 - x)^3
```

```
Out[1]= (1 - x)^3 (2 + 4 x^2)^2
```

`Expand` expands out products and powers, writing the polynomial as a simple sum of terms.

```
In[2]:= t = Expand[%]
```

```
Out[2]= 4 - 12 x + 28 x^2 - 52 x^3 + 64 x^4 - 64 x^5 +
        48 x^6 - 16 x^7
```

`Factor` performs complete factoring of the polynomial.

```
In[3]:= Factor[ t ]
```

```
Out[3]= -4 (-1 + x)^3 (1 + 2 x^2)^2
```

`FactorTerms` pulls out common factors from each term, but does not do complete factoring.

```
In[4]:= FactorTerms[ t ]
```

```
Out[4]= 4 (1 - 3 x + 7 x^2 - 13 x^3 + 16 x^4 - 16 x^5 +
        12 x^6 - 4 x^7)
```

There are several ways to write any polynomial. The functions `Expand`, `FactorTerms` and `Factor` give three common ways. `Expand` writes a polynomial as a simple sum of terms, with all products expanded out. `FactorTerms` pulls out common factors from each term. `Factor` does complete factoring, writing the polynomial as a product of terms, each as small as possible.

When you have a polynomial in more than one variable, you can put the polynomial in different forms by essentially choosing different variables to be “dominant”. `Collect[poly, x]` takes a polynomial in several variables and rewrites it as a sum

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of terms containing different powers of the “dominant variable” x .

Here is a polynomial in two variables.

```
In[5]:= Expand[ (1 + 2x + y)^3 ]
```

```
Out[5]= 1 + 6 x + 12 x^2 + 8 x^3 + 3 y + 12 x y + 12 x^2 y +
        3 y^2 + 6 x y^2 + y^3
```

Collect reorganizes the polynomial so that x is the “dominant variable”.

```
In[6]:= Collect[% ,x]
```

```
Out[6]= 1 + 8 x^3 + 3 y + 3 y^2 + y^3 + x^2 (12 + 12 y) +
        x (6 + 12 y + 6 y^2)
```

If you specify a list of variables, Collect will effectively write the expression as a polynomial in these variables.

```
In[7]:= Collect[ Expand[ (1 + x + 2y + 3z)^3 ], {x, y} ]
```

```
Out[7]= 1 + x^3 + 6 x^2 y + 12 x y^2 + 8 y^3 + 9 z + 27 z^2 +
        27 z^3 + x^2 (3 + 9 z) + x y (12 + 36 z) +
        y^2 (12 + 36 z) + x (3 + 18 z + 27 z^2) +
        y (6 + 36 z + 54 z^2)
```